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Topical Review

A review of next generation bifacial solar farms: predictive modeling of energy yield, economics, and reliability

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Abstract

Photovoltaic (PV) cell technology has made great progress over the past few decades, bringing the PV energy cost down to a point where it is competitive to conventional electricity prices. While monofacial panels have historically dominated the market, recent developments in the manufacturing of bifacial panels (collecting light from both faces) have made them accessible for commercial applications. It is therefore imperative to define the design principles so that the steeply expanding market of bifacial modules and PV farms stays on an efficient path. In this paper, we will discuss physics-based models for these next-generation bifacial PV farms for analyzing yield and costs. Besides the conventional farm configurations, tracking systems are gaining market shares aiming to enhance the yield at a lower cost. Within $\pm 30°$ latitudes, we predict a 20%-30% energy gain for fixed-tilt bifacial over monofacial modules and an additional 20%–40% gain for single-axis bifacial tracking. Compound systems such as agrophotovoltaics and floating PV applications may be the possible future for a sustainable merger of food-water-energy systems. We show a competing relation between active light collection on crops and energy yield in an agrophotovoltaics system vs panel density—the final design will be decided by the crop yield or light usage efficacy constraint. The output reliability in terms of soiling and module degradation is also explained in this paper. Solar farms are expected to see 2%–5% loss in revenue in Asia and the Middle East even after optimal cleaning. Additionally, the bifacial modules degrade ~0.5%–0.6%/year. While there have been several physics-based degradation analyses, the bifacial technology lacks a large enough data set of long-term degradation studies for accurate predictions. The combined economics of reliability against yield will decide the viability of the next generation bifacial PV industry.

Keywords: bifacial, solar farm, LCOE, reliability

(Some figures may appear in color only in the online journal)

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1. Introduction

The 6% efficient solar cells developed at Bell Labs in 1954 opened up a pathway towards a new industry for renewable energy. Later in the early 1960s, Shockley and Queisser defined that the potential limit for single-junction solar cells can reach up to 33%–34% efficiency under ideal situations [1, 2]. Yablonovitch *et al* showed that in practice, Si solar cells cannot go beyond 29.8% due to fundamental electronic losses and limited light trapping [3, 4]. These insights have helped the photovoltaics (PV) industry improve the silicon solar cell technology so that today it operates close to the fundamental limit.

With the improved efficiency and reduced PV module prices, global solar farm installations have been growing rapidly. In particular, the suitability of emerging cell design (e.g., PERC, PERT, HIT, etc.) with bifacial topologies has led to accelerated deployment of bifacial farms across the MiddleEast, Latin America, and Asia. Bifacials, which can inherently collect light through both its transparent faces, are now preferred for most new solar farm installations in MiddleEast and Latin America. Just like the works inspired by Shockley–Queisser and Yablonovitch guided the remarkable advancement in cell technology, we now need to understand the limits and losses in a solar farm and define the new organizing principles and relative merits of worldwide installation of fixed-tilt, tracking, floating, tandem, agro/aqua PV farms. In particular, as the bifacial module technology gains popularity, their thermodynamic [2] and practical limits related to the compound effects of light from sky and ground albedo need to be analyzed in detail. Towards that goal, several predictive models have been developed to analyze and design solar farms. Physics-based models mainly follow two different formulations: ray-tracing (RT) and view factor (VF). As the name suggests, RT follows the path for each ray (both direct and scattered) to evaluate the light collected by the panels. For example, Bifacial Radiance software from NREL [5] uses ray tracing. The VF method finds the integrated scattered light instead of tracing individual rays. This method uses the same principle as ray tracing while addressing direct light on panels or the ground. Several analyses on bifacial standalone panels [6] and their arrays [7] laid down the ground-work for the VF based PV farm models. PVsyst [8], NREL bifacial VF model [9], Purdue-bifacial model [10–12], etc. have been used for several studies to analyze local and global bifacial solar farm yield. Both the ray tracing and VF models have been shown to be accurate in predicting bifacial panel yields [13, 14]. While the farm yield is an important parameter, it is equally essential to determine the relevant economics and reliability. In the end, technical advancement, costs, and policies will determine the roadmap for the PV industry-driven energy sustainability [15, 16]. Figure 1 shows a host of attainable systems possible through the bifacial technology.

In this paper, we will discuss the end-to-end approach of the Purdue-bifacial model and the insights obtained by the use of models for world-wide installation of various types of solar farms. Section 2 describes the model inputs solar insolation (direct and diffuse) and predicts spatial and temporal light collection on the ground and on both faces of the panel. A coupled opto-thermo-electric model calculates the power output of the non-uniformly illuminated panel.

To design a solar farm, one would need to minimize the levelized cost of energy (LCOE) which involves costs related to installation, operation, and maintenance. In section 3, we explain how this analysis can be significantly simplified if we equivalently minimize for the essential LCOE (denoted by LCOE*). We predict a 20%–30% gain in yield from fixed-tilt bifacial farm compared to a conventional monofacial farm within $\pm 30°$ latitudes.

As discussed in section 4, external (soiling) and internal (module degradation) factors can lower the efficacy of the farm with time. We expect a 2%–5% loss in revenue in highly dust prone regions such as Asia and the Middle East even after optimal cleaning of the panel arrays. Soiling models predict the optimal cleaning cycle to minimize soiling losses. There have been several works on developing physics-based models for module degradation for use in energy yield and determining degradation pathways of existing farms [17–22]. While bifacial panels are relatively immune to potential induced degradation (PID), their overall degradation rate is comparable to monofacials [20]. A proper framework for module degradation helps us understand the physics so that we can redesign the modules appropriately tailored for each climatic condition.

Finally, in section 5, we discuss the emerging PV applications and how the numerical models can be adapted to analyze these systems. For example, our numerical analysis on bifacial tracking systems shows a possible 20%–40% gain in yield compared to fixed-tilt bifacial farms. We also discuss opportunities in agrophotovoltaics and floating PVs (FPVs). Both co-utilize the farm area and have the potential for the sustainability of food-energy-water.

2. Solar farm modeling

Panel yield in a solar farm depends on local conditions such as sunlight, temperature, humidity, wind speed, etc. The most important factors that dictate the output are time-varying irradiance, angle of incidence, and temperature. These can widely change depending upon the latitude along with other geographic and weather conditions. In this section, we will describe how we model the location-specific, and time-varying solar irradiance. The collection of light on the panels and the ground are numerically calculated, followed by the effects of temperature on panel yield. The electrical output of the panels will of course be affected by dust accumulation (i.e. soiling) and long-term panel degradation in any practical installation. A summary of this end-to-end Purdue-bifacial model is shown in figure 2. We will discuss the essentials of the model in the following subsections. Details can be found in prior literature [10–12].

2.1. Irradiance and sunpath

The temporal intensity and position of the sun is the input to the solar farm calculation model. The estimated farm yield will depend on the location-specific input dataset. In our numerical

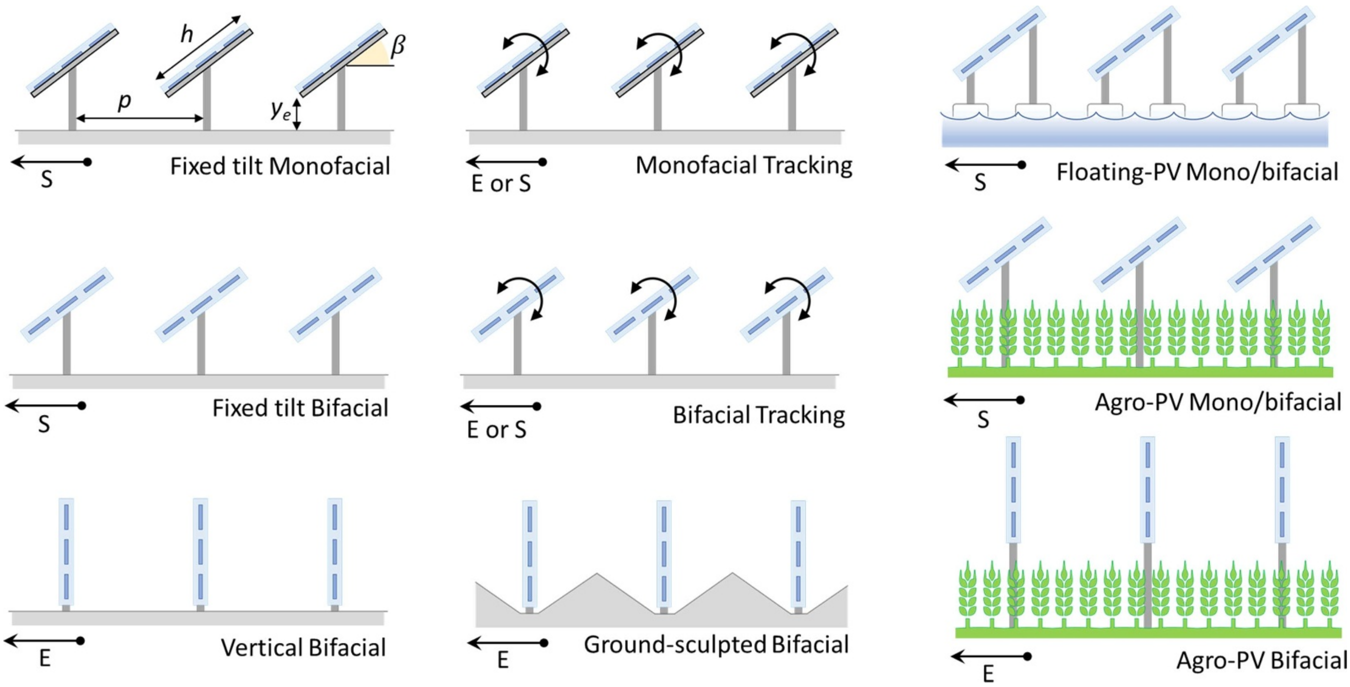

Figure 1. Bifacial modules have broadened the design options for solar farm design and the need to optimize them for location-specific illumination and degradation conditions.

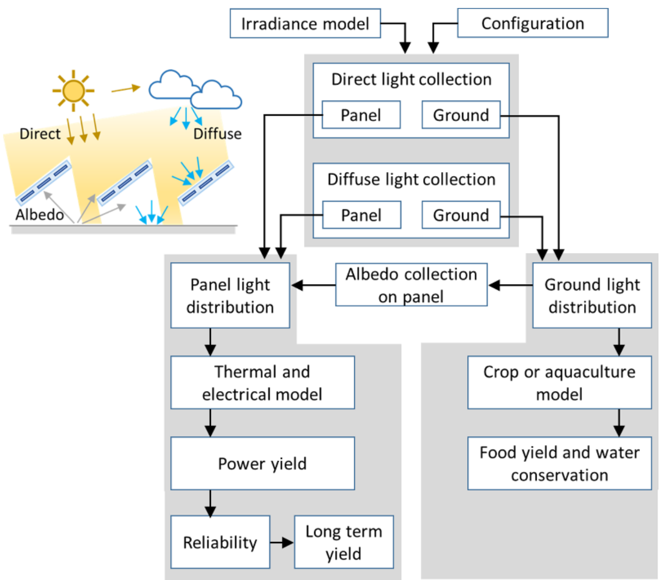

Figure 2. The Purdue-bifacial model allows a comprehensive end-to-end optical-electrical-thermal-cost optimization of solar farms.

models, we use NREL's solar position method implemented within Sandia's PVLib model. For a specific location, this gives the sun-path defined by the time-varying zenith angle (θ_Z) and azimuth angle (γ_S) of the sun. Haurwitz clear sky model is scaled to more practical, location-specific insolation values from NASA POWER database to estimate global horizontal irradiance (GHI, I_{GHI}) for each time-step. Finally, the GHI is decomposed into direct (DNI, I_{DNI}) and diffuse light (DHI, I_{DHI}) following Orgill–Hollands [23], and Perez models [24].

In the solar farm output calculations, DNI, DHI, and the sun-path are considered as inputs. Therefore, in principle, one could collect DNI and DHI data from other sources. For example, there are many other techniques to split GHI [25, 26]. There are also various weather stations across the globe that provide data on DNI and DHI—this could ideally result in better output estimates.

2.2. Light collection and output of panels

While the location (i.e. latitude and longitude) determines the ambient input conditions of a solar farm, the light collection will also depend on the configuration (as marked in the top-left setup in figure 1): orientation or azimuth γ_A, panel tilt β, panel width h, row period p, and panel elevation y_e. We consider infinitely long rows—this reduces the system to a 2D problem as shown in figure 1. The analysis can be done within a single period as we also assume infinite number of rows or periods. For a practical, finite-size solar farm, the edge effect would need further analysis.

Solar panels (both monofacial and bifacial) collect direct and diffuse light from the sky. The reflected diffuse light from the ground (albedo light) can however be collected most effectively only with the bifacial panels. The bifacial panel output would therefore depend on the albedo R_A. To model this system, we first calculate the light intensity distribution $I_{gnd}(x)$ on the ground (combined effect of DNI and DHI) as formulated in table 1. Here $F_{dx-sky}(x)$ is the integrated VF from the ground at x to the periodically unobstructed sky seen between the rows [12, 27]. The light reflected from the ground $R_A I_{gnd}(x)$ now acts as a secondary source for the panels.

Next, we calculate light collected at any position l on the panel (both from the sky and the ground). We only show the

Table 1. Light on the ground.

For any position x on the ground:

$$I_{\text{gnd:DNI}}(x) = \begin{cases} I_{\text{DNI}} \cos\theta_Z & ; x \text{ is not shaded} \\ 0 & ; x \text{ is shaded} \end{cases} \quad (1)$$

$$I_{\text{gnd:DHI}}(x) = I_{\text{DHI}} \times F_{\text{dx}-\text{sky}}(x) \quad (2)$$

$$I_{\text{gnd}}(x) = I_{\text{gnd:DNI}}(x) + I_{\text{gnd:DHI}}(x) \quad (3)$$

Table 2. Light on the panel.

For any position l on the front of the panel:

(a) Light collection from the sky:

$$I^{\text{F}}_{\text{PV:DNI}}(l) = \quad (4)$$

$$\begin{cases} I_{\text{DNI}} \cos\theta_{\text{F}} \left(1 - R(\theta_{\text{F}})\right) \eta_{\text{dir}} & ; l \text{ is not shaded} \\ 0 & ; l \text{ is shaded} \end{cases}$$

(θ_{F} is the angle of incidence)

$$I^{\text{F}}_{\text{PV:DHI}}(l) = I_{\text{DHI}} \times F^{\text{F}}_{dl-\text{sky}}(l) \times \eta_{\text{diff}} \quad (5)$$

(b) Light collection from the ground:

$$I^{\text{F}}_{\text{PV:Alb}}(l) = \eta_{\text{diff}} \int_0^p R_{\text{A}}\, I_{\text{gnd}}(x) \times F^{\text{F}}_{dl-\text{dx}}(l,x)\, \text{d}x \quad (6)$$

(c) Net light collection on the panel:

$$I^{\text{F}}_{\text{PV}}(l) = I^{\text{F}}_{\text{PV:DNI}}(l) + I^{\text{F}}_{\text{PV:DHI}}(l) + I^{\text{F}}_{\text{PV:Alb}}(l) \quad (7)$$

$$I_{\text{PV}}(l) = I^{\text{F}}_{\text{PV}}(l) + I^{\text{B}}_{\text{PV}}(l) \quad (8)$$

Table 3. Thermal model.

$$\eta(T_{\text{cell}}) = \eta_{\text{STC}}\left(1 - \text{TC}\left(T_{\text{cell}} - T_{\text{STC}}\right)\right) \quad (9)$$

$$T_{\text{cell}} = T_{\text{amb}} + c_{\text{T}} \frac{\tau\alpha}{u_{\text{L}}} P_{\text{POA}} \left(1 - \frac{\eta(T_{\text{cell}})}{\tau\alpha}\right) \quad (10)$$

$$P_{\text{POA}} = P_{\text{PV:DNI}}/\eta_{\text{dir}} + \left(P_{\text{PV:DHI}} + P_{\text{PV:Alb}}\right)/\eta_{\text{diff}} \quad (11)$$

here, $P_{\text{PV:\#}} = \int \left(I^{\text{F}}_{\text{PV:\#}} + I^{\text{B}}_{\text{PV:\#}}\right) \text{d}l$

$$P_{\text{PV:T}} = \eta(T_{\text{cell}}) \times P_{\text{POA}} \quad (12)$$

formulation for light collection on the front face of the bifacial panel in table 2—the analysis for the back face follows similarly. The direct and diffuse light from the sky and the albedo light collected on the panel can be calculated as shown in table 2. The collected light is then multiplied by the power conversion efficiency of the panel to get the electrical output. It is important to note that all these calculations are done for each time-step (e.g. every 1 min). The light collection distribution $I_{\text{PV}}(l)$ on the panel is used with an electrical circuit model (with N-bypass diodes) to find the power output $P_{\text{PV}}(t)$.

2.3. Temperature correction

The solar cell efficiency can decrease linearly with the operating temperature T_{cell} as shown in equation (9). T_{cell} is determined by the total plane of array (POA) collected light P_{POA}. Table 3 summarizes Skoplaski's model [28, 29] to estimate the temperature corrected panel efficiency $\eta(T_{\text{cell}})$ and output $P_{\text{PV:T}}(t)$. Integration of $P_{\text{PV:T}}(t)$ over 24 h gives the energy output $E_0(n)$ for nth day of the year.

Bifacial panels have a lower temperature coefficient TC compared to monofacials. Typical values of the parameters in the Skoplaski's model for a 16.8% (η_{STC}) Si Heterojunction bifacial solar module are [12]: TC $= 0.26\%/^\circ$C, heat loss coefficient $u_{\text{L}} = 21.5$, temperature and irradiance averaging correction $c_{\text{T}} \approx 1$, $\tau = 0.95$ and $\alpha = 0.8$. Monofacial solar modules (e.g. Al-BSF) have higher TC ($0.41\%/^\circ$C).

2.4. Soiling

Thus far, we have considered the model for a solar farm comprised of clean panels. During operations, the panels in the farm can get soiled from dust, snow, or bird droppings. Snow and bird droppings are usually irregular and are difficult to analyze. However, dust accumulation is common to all PV installations and requires periodic cleaning. The soiling effect can be represented by an average soiling loss a (in %-loss/day). The yield of the soiling affected solar farm on the nth day of the year can be written as:

$$E(n) = E_0(n) \times f_{\text{L}}(n, n_{\text{u}}). \quad (13)$$

Here, $f_{\text{L}} = \exp(-a\, n_{\text{u}})$ is the relative yield n_{u} days after cleaning. If we assume that the panels are periodically cleaned every n_{C}-days, then on the nth day of the year, $n_{\text{u}} = (n \text{ modulo } n_{\text{C}})$. Immediately after cleaning, $f_{\text{L}} = 1$, and $E(n) = E_0(n)$ on that day.

2.5. Yearly yield

Finally, the daily yield can be summed over the days to obtain the yearly yield:

$$YY(p,\ \beta,\ h,\ y_e,\ \gamma_A,\ R_A) = \sum_{n=1}^{365} E(n). \quad (14)$$

As the farm continues to operate, the panel efficiency η_{STC} will degrade over time lowering YY in the consecutive years. We will discuss the analysis on panel degradation later in section 4.

3. Analysis: design and yield

Now that we know how to predict the yield of a solar farm for a set configuration, we would want to study the following: (a) What is the optimal design of a solar farm? (b) How do different configurations compare? (c) How does the output degrade over the years of operation?

3.1. Design criteria

The best design of a solar farm to maximize yield per land area is trivial: horizontally cover the complete farm area with panels (i.e. zero spacing in between). In that case, we can collect the full GHI to achieve the maximum yield. This, however, may not be the most economic solution. In practice, we would aim to minimize the levelized cost of energy (LCOE). LCOE is defined as the ratio of the total cost to total energy yield. Total cost includes land and module prices, installation, and maintenance costs. Instead of minimizing LCOE, one can minimize the 'essential LCOE' defined as follows [11]:

$$\text{LCOE}^* = \frac{M_{\text{L}} + p/h}{\text{YY}} \quad (15)$$

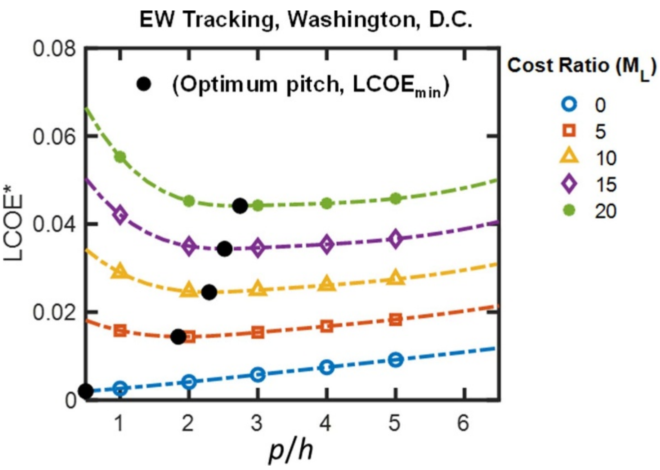

Figure 3. LCOE* as functions of p/h and M_L are mapped here for EW tracking bifacial solar farm. The optimum design p/h is marked as dark circle for each M_L. Reprinted from [30], Copyright (2021), with permission from Elsevier.

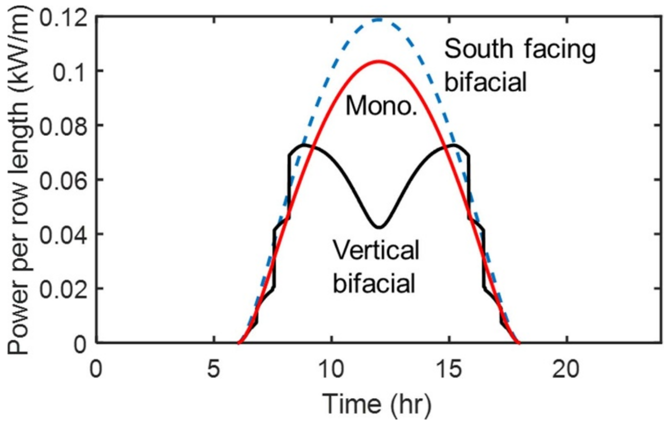

Figure 4. Power per panel shown for south facing mono- and bifacial array, and EW facing vBF array. Bypass diode switching is apparent from the jumps in the plot for vertical bifacial panel. Although vertical configuration shows lower output per panel, it has relatively improved output per land area as it is more densely packed.

where, M_L is the ratio of the module to land costs (converted to net present values). The value of M_L will decide the optimum configuration (i.e. p/h, β, and γ_A) of the farm. For example, LCOE* values within the (p/h, M_L)-design space and the optimum p/h for different M_L for east–west (EW) tracking bifacial solar farms are shown in figure 3 [30]. Originally proposed by the Purdue group [11], the concept of essential LCOE (i.e. LCOE*) decouples economic factors such as capital and maintenance costs from the farm design step, thereby greatly simplifying the optimization process. The only economic factor required is the module to land cost ratio M_L.

3.2. Understanding the output

In this section, we will focus on the yield of a clean panel array (no soiling). We will discuss the effects of soiling in the next section. An example case of temporal power output $P_{PV:T}(t)$ is shown in figure 4 for Washington DC in September. The south-facing bifacial (sBF) solar panel array shows the typical bell-shaped output with a peak at the solar noon (blue dashed line). The EW facing vertical bifacial (vBF) solar farm shows its characteristic double-humped output (black solid line). As the sun is vertically above the ground at noon, the vertical panels do not receive direct sunlight. The panels will only receive DHI and albedo light, resulting in a dip in output at noon. Therefore, there are peaks a few hours before and after noon. The sudden jumps or discontinuity on the output for vBF (black line) is due to the three-bypass diodes turning on/off as the rows of panels are mutually shaded in early mornings or late afternoons.

We obtain the yearly energy yield YY by integrating these power outputs. For the sBF, LCOE* is minimized by calculating YY for various (p/h, β) combinations at a set M_L. Figure 5 presents the monthly output breakdown of sBF. These have been optimized separately at each location for $M_L = 12$. Washington DC and Sydney being in the northern and southern hemispheres have energy peaks around June–July and December–January respectively.

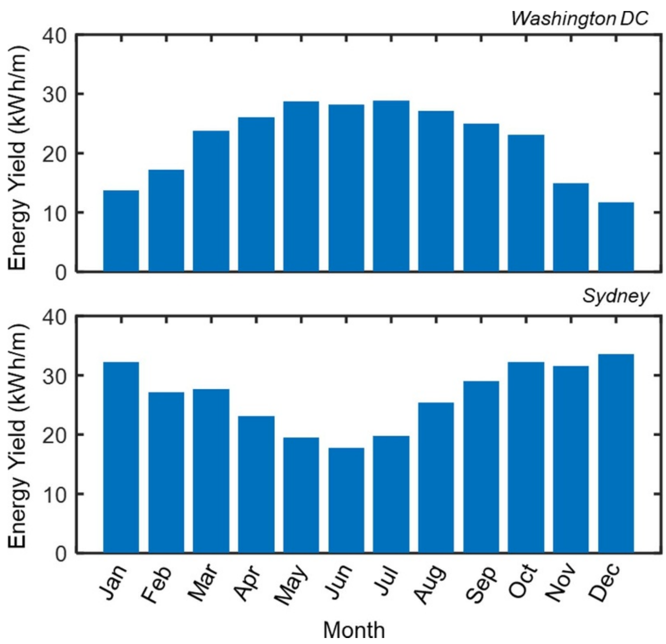

Figure 5. Monthly energy yield (per row length) is shown for south facing bifacial solar farm. The yield peaks in June–July for northern hemisphere (Washington, DC), and in December–January for southern hemisphere (Sydney).

These calculation steps can be repeated for all locations across the globe to form a predictive yield map of any configuration.

3.2.1. M_L determines LCOE and optimal farm configuration.

Figure 6 shows %-reduction in LCOE* for optimized bifacial vs monofacial solar farm [11]. When the module cost is negligible compared to the land ($M_L \approx 0$), we would pack the modules as close to each other as possible. This leaves very small room for bifacial gain (see figure 6(a)) as very little light would be collected at the back surface. At higher M_L, the modules are

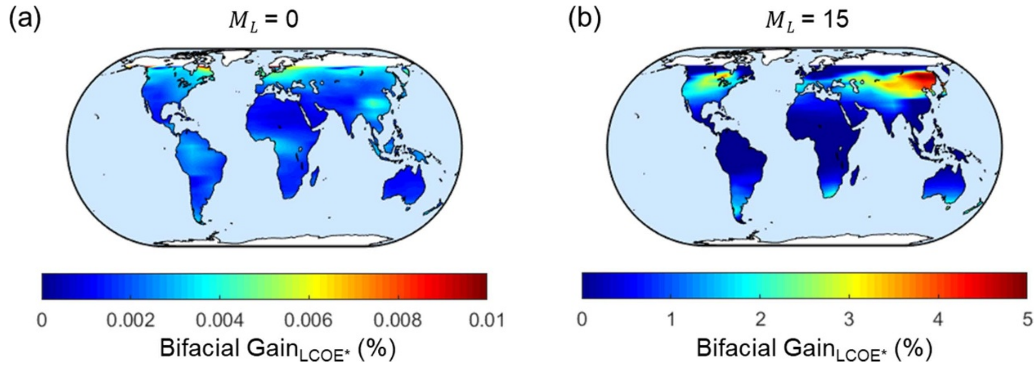

Figure 6. The optimal design and bifacial gains are affected by the module to land cost ratio M_L. For low module cost ($M_L = 0$), we would closely pack the modules horizontally leaving no room for light collection at the back face. This gives very low bifacial gain. When $M_L = 15$, the panels are spaced and correspondingly tilted. In this case, we see a 2%–5% bifacial gain in LCOE* (i.e. reduction in LCOE*) above 30° latitude. Reprinted from [11], Copyright (2019), with permission from Elsevier.

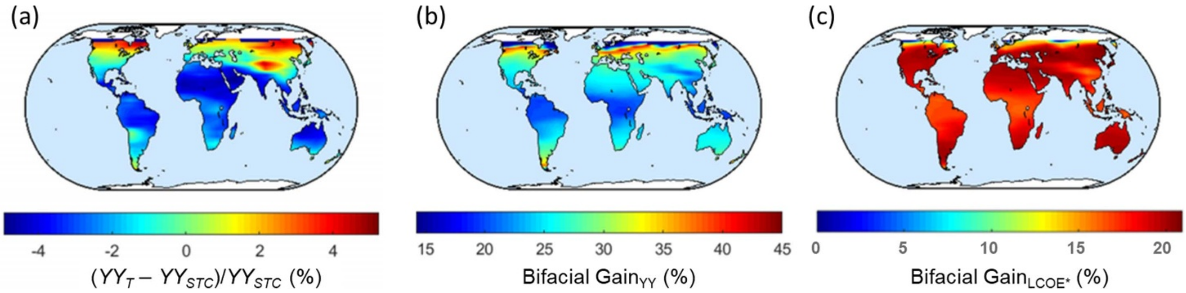

Figure 7. Increased temperature lowers solar cell performance assuming $M_L = 15$. (a) With temperature corrections, we see lower (or higher) yearly yield (YY_T) compared to that in STC for hot (or cool) climates. (b) Bifacial panels are less sensitive to temperature compared to monofacials. Therefore, bifacial gain in YY is more pronounced once the temperature effects are considered—this is a more accurate representation of what is to be expected in practice. (c) This map shows the corresponding bifacial gain in LCOE*. Reprinted from [12], Copyright (2020), with permission from Elsevier.

further spaced and tilted optimally towards the sun, leaving room for DHI and albedo collection. Bifacial gain in LCOE* is therefore more at higher M_L. In practice, there will always be a minimum row-to-row spacing required for maintenance. Therefore, with a minimum row-spacing constraint, there will be a bifacial gain even if the advancements in manufacturing technology results in $M_L \approx 0$.

3.2.2. *Temperature correction changes YY estimate by ±5%.*

As discussed earlier, PV efficiency decreases at higher temperatures. Compared to YY estimated at STC for Si-HJ bifacial farm, the temperature corrected YY, therefore, is lower in hot climates and higher in cooler climates as shown in figure 7(a) [12]. Without temperature correction, YY accuracy can vary upto ±5%. Besides, bifacial panels have lower temperature coefficient TC compared to monofacials. This added benefit in bifacial yield can only be analyzed through a temperature-aware solar farm model.

Finally, figures 7(b) and (c) show the bifacial gain in YY and LCOE* of Si-HJ bifacial farm compared to Al-BSF monofacial farms (for $M_L = 15$). The gain in YY is ~20%–30% close to the sun-belt, and even higher at higher latitude.

3.2.3. *The design of vertical solar farms needs nontrivial optimization.*

EW-facing vertically aligned panel array has a few key advantages over the south-facing tilted panel arrays. First, vertical surfaces inherently accumulate less dust compared to tilted ones. And, more panels can be packed within a farm area even after leaving space for maintenance compared to an array of tilted panels. As we will discuss in the following paragraphs, at many locations, the denser vertical panel array can give more output per farm area.

Vertical bifacial solar farms may not always be promising compared to monofacial panels [10]. However, if we pattern the ground appropriately, the albedo light collection can be enhanced [31]. The ratio of yearly yield per land area between ground sculpted vertical bifacial farm (GvBF) and optimally tilted monofacial farm is mapped in figure 8. We see that GvBF has a higher yield for latitudes > 20°. Close to 40° latitude, the gain is more than 30%. This analysis assumes 10% soiling loss for the monofacial farm.

We note that GvBF shows output gain only when the yield is estimated *per land area*. This definition is mainly relevant when M_L is small. For higher M_L (e.g. ~12), the bifacial gain in terms of LCOE* should be analyzed. In that case, the economic advantage of GvBF diminishes—after all densely packing panels vertically will increase the capital cost as we are essentially using more panels on the same area of land.

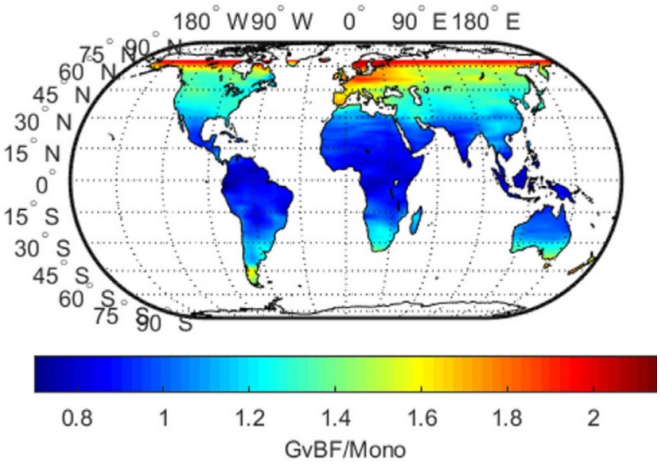

Figure 8. A world-wide yearly yield ratio of optimal ground sculpted GvBF to optimal south-facing monofacial solar farm shows that GvBF may be beneficial for latitudes beyond 30°. Reprinted from [31], Copyright (2019), with permission from Elsevier.

4. Yield degradation

The output of the solar panels in a farm may degrade over time due to both external and internal conditions. Externally, dust is expected to accumulate gradually lowering the output. This of course can be reset by cleaning the panels. Changes and degradation within the module cannot be recovered in most cases.

4.1. Soiling effects and cleaning cycles

The daily panel yield $E(n)$ will decrease compared to the estimated clean-panel yield $E_0(n)$ until the panel is cleaned, as described by equation (1). In a PV system, we would generally clean the panels every n_C days. To maximize the yield, we would need to clean the panels every day. This will however decrease the revenue due to the recurring cleaning cost. If we leave the panels uncleaned for too long then the overall energy yield will decrease thereby lowering the revenue. It is therefore expected that there is an optimum cleaning interval for which the revenue and profit will be maximized. Instead of maximizing the revenue or profit, we can equivalently maximize the 'normalized revenue'. The normalized revenue is the ratio of the net cash inflow to the clean-farm revenue:

$$f_R = \frac{RE_S - C}{RE_{S0}} \tag{16}$$

here, $E_S = \sum E(n)$ is the net energy yield summed over a cleaning interval of n_C days. Correspondingly, $E_{S0} = \sum E_0(n)$ is the anticipated yield of a soiling-free farm over the same n_C days. Also, C is the cost per cleaning of the farm, and R is the electricity tariff.

For example, we compare south-facing monofacial panel array tilted horizontally and at 24°, and EW-facing GvBF array [32]. The horizontal panel has the highest, and the vertical panel has the lowest soiling rate, as seen from the slopes in figure 9(a). The optimum cleaning cycles n_C^{opt} of these configurations are decided from the maximum f_R condition in figure 9(b). Although this gives us n_C^{opt}, the normalized revenues f_R of the three setups do not portray the relative economic benefits as the rated, clean-farm yields E_{S0} are unequal. For appropriate comparison, we should study the revenues as shown in figure 9(c). If we choose to clean each of the configurations at optimal intervals (i.e. peak position on the revenue plot), the 24° tilted south-facing panel would give the highest revenue. However, at PV sites with difficult accessibility (e.g. agrophotovoltaics installations), if we require to clean the panels after 25–30 d, the vertical panel array would be a better option. A global perspective on optimal normalized revenue is mapped in figure 10 [33] for tariff rate of 0.069 USD kWh^{-1}, cleaning cost of 0.027 USD kW^{-1}, and location-specific GHI and soiling rates based on 100 MW farm estimates. Of course, the specific results depend on the array period, cleaning cost, electricity tariff, etc.

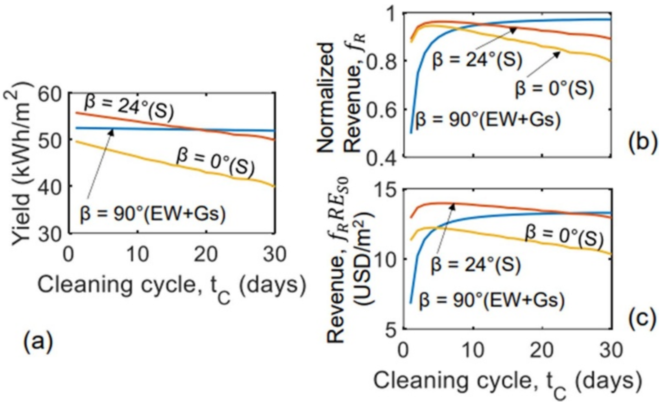

Figure 9. (a) Energy output of soiling affected farms integrated over November–February (4 months) in Dhaka for varying cleaning cycle t_C for the three different configurations. Higher tilt shows lower soiling loss rate. Normalized revenue and the net 4 months revenue shown in (b) and (c). © [2020] IEEE. Reprinted, with permission, from [32].

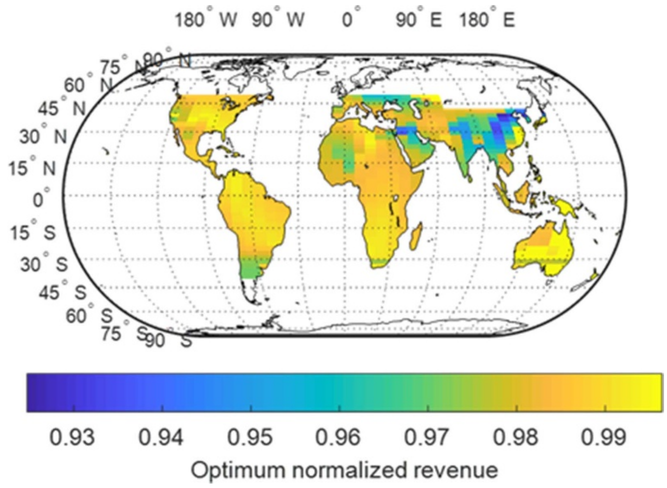

Figure 10. A world-wide prediction of optimal normalized revenue assuming tariff rate of 0.069 USD kWh^{-1}, cleaning cost of 0.027 USD kW^{-1}, and location specific GHI and soiling rates. Reprinted from [33], Copyright (2021), with permission from Elsevier.

4.2. Module degradation and reliability

The efficiencies of in-field modules will degrade over time. Although LCOE* allows us to optimize farm design, it does not account for the degradation of the modules in the following years. This is, however, an important consideration addressed through LCOE while evaluating the farm economics. The LCOE is related to LCOE* through the following [11]:

$$\mathrm{LCOE} = \mathrm{LCOE}^{*}\,\frac{\mathbb{C}_L}{\chi} \tag{17}$$

where, $\mathbb{C}_\mathrm{L}$ is the net present value of the land cost, and χ is a correction factor defined over the module lifetime (Y) as:

$$\chi = \sum_{k=1}^{Y} (1-d)^k (1+r)^{-k}. \tag{18}$$

Here, r and d are the discount rate and the module degradation rate, respectively. The land cost $\mathbb{C}_L$ depends on r. And, the effect of module degradation is included in the LCOE calculations through the factor $\chi(r,d)$.

The module efficiency degrades over time due to yellowing, corrosion, solder-bond failure, PID, and so on. Although bifacial technology dates back to the 1970s [34], its increasing popularity as a preferred solar cell technology for large-scale solar farms is a relatively new phenomenon [35]. As a result, while the module level reliability of bifacial solar cells is well-established [36], most of the published field data rely on results from relatively small systems [20, 37]. It will be important to analyze the large scale field-data to guarantee the viability of the technology.

There are important structural differences between monofacial vs bifacial modules. In particular, unlike the opaque back sheet of monofacial solar cells, the bifacial solar cells use glass or transparent plastic back sheets. While monofacial metal contacts cover the back of the silicon solar cells, the grid on the back of bifacial solar cells is optimized for light transmission and contact resistance [38].

The structural difference allows a significant fraction of the sub-bandgap photons to transmit through a bifacial cell, with the corresponding reduction in the module temperature and reliability improvement [39]. Moreover, the reduced temperature coefficient associated with higher operating voltage (V_mp) in HIT or back-contact passivated cells [2, 40] ensures higher energy yield compared to monofacial cells.

A large-scale study in 2016 has shown that monofacial x-Si (high-quality crystalline silicon) based solar cells (both bifacial and monofacial) have a degradation rate of approximately 0.5%–0.6%/year, while the silicon heterojunction cells have a higher degradation rate of approximately 1%/year [37, 41]. In contrast, a recent study analyzing the degradation of 10 year-long bifacial modules [20] finds that the bifacial module losses to be comparable to monofacial loss. The key mode of degradation appears to be the increase in series resistance (possibly in the junction box, because extensive module-level EL, PL, and thermal imaging characterization did not find any hot-spots associated with solder-bond failure). The characteristic nonlinear decrease in V_OC has been traced back to the increase in the dark current due to reduced minority carrier lifetime. Bifacial solar cells have essentially similar PID degradation as in monofacials, with very little shunt degradation in the fielded modules [20] and laboratory experiments [42]. In fact, it has been demonstrated that backside PID is reversible [36] if it has not been caused by the local corrosion of Si at the rear side [43].

At the large-scale farm-level, several emerging software tools will simplify the analysis of time-series data from the field. Two classes of techniques have been proposed. The statistical techniques in the RdTools use clear-sky normalization, Year-over-year nonlinear degradation calculation, etc [44, 45]. Further generalization of the technique based on an unsupervised machine learning approach has been proposed. The approach relies only on the measured power signal as an input and a generalization of the principal component analysis to extract daily/yearly time-periodic signals [46]. The second class of physics-based online characterization technique augments the statistical analysis by explicitly correlating the power-degradation information to the underlying degradation mechanisms. For example, the real-time resistance method [47] can directly determine the changes in the series resistance as a proxy for solder-bond failure, the analytic I_sc–V_oc method [48] characterizes the degradation in terms of uniform current loss, recombination loss, series resistance loss, and current mismatch loss, etc, and finally, the most general approach involving suns-V_mp method [21, 49] creates a pseudo I–V curves to track the five-parameters associated with the x-Si solar cells and correlated to the corresponding degradation modes. Extensive use of these statistical and physical techniques are expected to generate a wealth of information for bifacial solar farms currently being installed worldwide.

5. Emerging applications of bifacial solar farms

5.1. Solar tracking

Instead of fixing the panel tilt at a certain angle, we could continuously move the panel so that we can maximize the energy yield. Such solar tracking can follow the sun throughout the day using a 2-axis solar tracking system. A 1-axis tracking system, on the other hand, would follow the sun from east to west (EW-tracking), but cannot move along north or south. Another variation of 1-axis tracking can be thought of as a fixed tilt panel that can move to correctly follow the seasonal sun-path (NS-tracking). These systems require mechanically moving parts and would add to the cost. Therefore, the 1-axis system is more popular compared to the 2-axis tracker. According to the report from the International Technology Roadmap for Photovoltaic (ITRPV), 1-axis tracking PV shares ~30% of the total PV market in 2020 [35]. This share is projected to go beyond 40% by 2030.

The introduction of 1-axis tracking in a monofacial panel array can increase the yield by 18% in locations near the equator [50]. Another work by Janssen *et al* showed that the 1-axis tracking monofacial and bifacial systems can have 15% and 26% gain respectively [51]. A separate work by

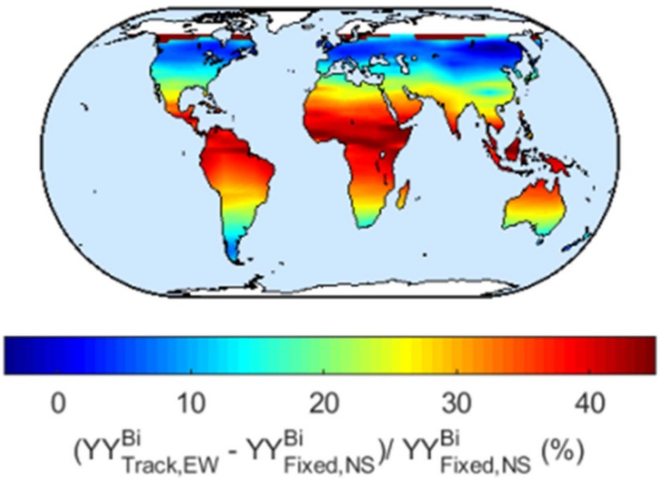

Figure 11. World-wide gain in yearly yield of EW tracking bifacial PV system vs fixed tilt conventional (north or south facing) bifacial PV farm. The gain can be over 40% near the equator. Reprinted from [30], Copyright (2021), with permission from Elsevier.

Rodriguez-Gallegos *et al* predicts ∼35% gain from a bifacial 1-axis tracking system. This system is therefore the cheapest PV configuration with 16% lower LCOE compared to conventional configuration [52].

Our model described in the previous sections, of course, can optimize and predict the output of EW or north–south (NS) 1-axis tracking systems. In the model, the tilt of the panel is time-varying $\beta(t)$ such that the output power of the panel is maximized at each time step. The EW-tracking (i.e. tracked hourly or continuously) system yield more than the NS-tracking (i.e. tracked seasonally) system. As shown in the map in figure 11, compared to fixed-tilt bifacial farm, the bifacial EW-tracking yields show >20% gain within the sun-belt regions. The gain can be as high as 40% near the equator.

In practice, for site-specific analysis of a tracking system, other works in the literature consider details such as partial albedo blocking due to the fixture and torque tubes [53]. These analyses have helped redesign the tracking fixtures to cut down these losses (Nextracker in bifiPV workshop 2020 [54]). In our analysis, we do not consider these losses and focus on other pertaining physics: e.g. tracking and bifacial gain, effects of location, temperature and installation height, etc.

5.2. Solar farms based on tandem solar cells

Multi-junction tandem cells are more efficient than single-junctions cells—this translates to enhanced yield from the same sized module. The tandem cell technology, however, has not been adopted for large-area solar farms due to its significantly higher costs. Recent developments in perovskite (PVK) solar cells have assisted the PVK-Si tandem technology to improve significantly. This is cheaper than the conventional III–V material based tandems. With a more stable and reliable PVK-Si tandem, this has the potential to be used in large area farms.

The group at Purdue was one of the first to propose bifacial PVK-Si tandem cells in 2015 [55]. The bandgap (E_G) of the perovskites (1.55 eV) at that time was mismatched with Si (1.12 eV) for a monofacial tandem—the bifacial tandem, however, showed improvements over a single junction mono and bifacial Si cell. Further developments in perovskite fabrication allowed researchers to tune the bandgap to appropriately form a tandem on a Si subcell [56, 57]. The perovskite bandgap would require separate tuning for different albedo conditions. Further, ∼30% bifacial gain has been predicted from three- and four-terminal bifacial PVK-Si tandems [58, 59]. Compared to conventional two-terminal tandems, these three- and four- terminal devices have the added benefit that the mismatched current can be recovered when the albedo light collection changes.

The bifacial solar farm modeling framework discussed in section 2 can be expanded to analyze and predict PVK-Si tandem panel arrays. The only change needed is in the electrical model: we would need to separately model the tandem cell and module efficiency as a function of front and backlight collection.

In the end, the prospect of PVK-tandem successfully entering the PV market greatly depends on the LCOE and reliability. The LCOE, which has its bottleneck at the combined effect of perovskite efficiency and manufacturing cost is estimated to have reached a point where PVK-tandem is competitive [59]. And, PVK-Si combination may enhance the perovskite layer lifetime as follows: (a) tandem cells produce more output resulting in lower self-heating, and (b) Si subcell layer is much thicker than the perovskite layer thereby retaining most of the heat of the device. Therefore, the perovskite cell is expected to be less heat-stressed in a tandem compared to single-junction modules. Finally, one must carefully redesign the internal electrodes for tandem cells, because the traditional laser-scribed thin-film solar modules have a different grid topology compared to busbar/finger based c-Si solar cell technology. Recent designs of Si-PVK tandem cells have used busbar/finger based design for both c-Si and perovskite cells.

5.3. Agrophotovoltaics

Agrophotovoltaic systems provide a unique opportunity to cultivate vegetation under the solar modules [60–62]. This approach does not only enable a dual land use for food-energy production but can also provide many symbiotic benefits for the food-energy-water nexus including reduced water budget and resilience against climate stress such as heatwaves and drought [63]. Indeed such systems could be highly valuable for hot and arid climates to sustain the future food-energy needs of the growing world population under the increasing effects of climate change.

Bifacial PV has opened new options for farm design and may significantly reduce the LCOE by reducing the height of the supporting structure. The design of an agrophotovoltaic farm requires an optimization of the spatial-temporal sharing of sunlight between solar modules and the vegetation underneath [64]. The modeling framework described in section 2 can be readily applied to calculate the spatio-temporal shading patterns over the crops for both direct and diffused sunlight

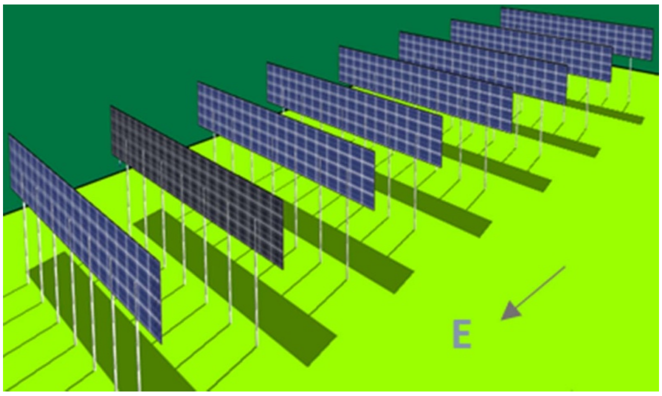

Figure 12. EW vBF solar arrays for an agrophotovoltaics farm. © [2020] IEEE. Reprinted, with permission, from [27].

along with the resultant effect on the albedo collection on the modules.

An interesting comparison for the fixed-tilt agrophotovoltaic module technology and the farm topology has been made between the traditional NS faced tilted array configuration with the EW faced vertical bifacial solar arrays [27]. The EW facing vertical panel array shown in figure 12 is more suitable over crops due to its relatively uniform integrated light distribution on the ground. Figure 13 shows the PV energy output and the normalized photosynthetically active radiation (PAR_r) available to crops for the two farm configurations as a function of row to row panel density. The general trends for the PV energy and PAR_r as a function of p/h are as expected: i.e. higher panel density increases the energy production at the cost of reduced PAR_r for the crops. Interestingly, the quantitative difference between the results for the two PV configurations is not significantly different when $p/h = 4$ which is about half of the standard density commonly used in PV farms. Many practical agrophotovoltaic farms prefer to use half density to ensure that crop yields are not substantially affected due to the reduction in PAR_r. For such systems, vertical bifacial technology may be attractive due to its additional benefits such as minimal ground coverage, lower elevations above ground that reduces installation costs, and easy mobility for farm machinery [65]. Moreover, as mentioned previously, the vertical installation has minimal soiling losses which can make it attractive for agrophotovoltaic especially when the dust is high (e.g. during the harvesting and tillage stages).

The modeling approach further allowed to define a figure of merit for the design of agrophotovoltaic solar arrays based on the relative PV energy production and PAR availability to the crops. LPF was introduced in [27], which estimates the efficacy of the solar array design by adding the relative yields for PV energy and the transmitted PAR at the crop level. Figure 13(b) shows that the best LPF can be obtained for the bifacial N/S faced agrophotovoltaic farm. Although LPF can be increased by increasing the array density, the relative PAR at $p/h \leqslant 2$ might be unacceptable for some crops. An optimal agrophotovoltaic farm design hence requires a careful balance between the sunlight requirement for the crop and the energy production which subsequently imposes practical limits to the

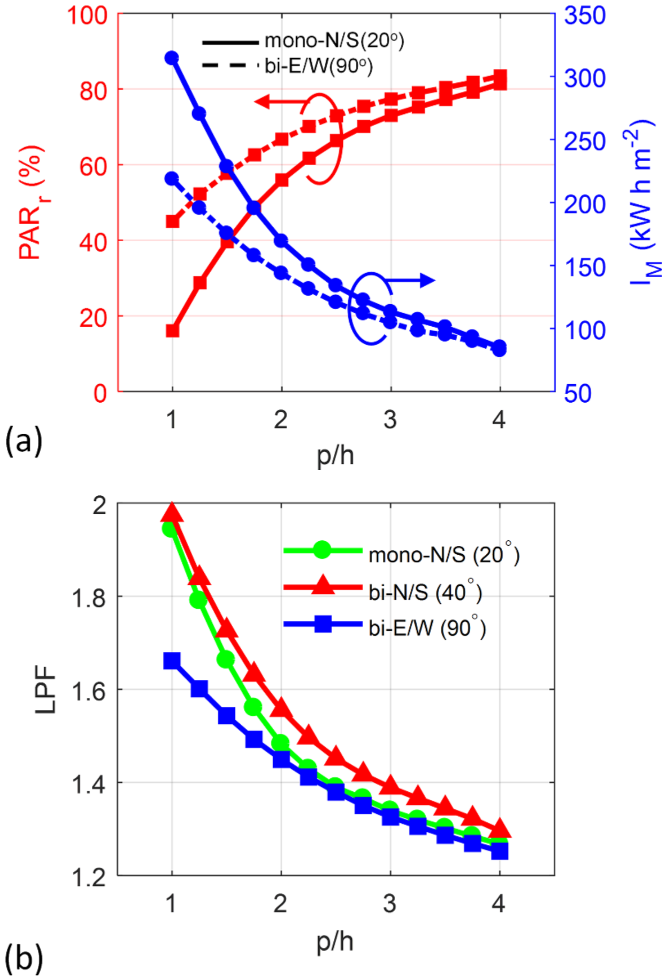

Figure 13. (a) The relative PAR and solar PV energy generation (I_M) for monofacial NS faced fixed tilt PV arrays vs that for the E/W faced vertical bifacial arrays. (b) Light productivity factor (LPF) for monofacial/bifacial N/S faced fixed tilt vs E/W faced vertical bifacial arrays. © [2020] IEEE. Reprinted, with permission, from [27].

array density and the LPF. Future research on this topic will include the application of the modeling framework on mobile panels which allow dynamic optimization of the tilt angle to obtain the best spatio-temporal sharing of sunlight between panels and the crops.

5.4. Floating PV (FPV)

FPV is relatively new—one of the initial FPV installations was done in 2007 [66] aiming to exploit the cooling effect of water to improve the operating efficiency of the panel. This is of particular interest where there are limited land areas usable for solar farm installation. While FPV currently covers less than 5% of the global PV market, it is projected to reach 10% by 2030 [35]. Besides the possibility of installation on large lakes, rivers, or coastal areas, FPV can also be designed for panels over fish farms to form a water-food-energy nexus [67].

Although the reflectance of water is low ($< 10\%$), the effective albedo of a water surface with ripples or waves can be higher. As a result, Tina *et al* has shown that bifacial FPV can produce up to 13.5% more energy compared to monofacial FPV [68]. Being 3 °C–4 °C cooler, monofacial FPVs will have better yield compared to PV arrays on the ground. However, the combined effect of lower albedo and lower temperature on bifacial FPV output compared to bifacials on land is yet to be quantified. Such analysis with effectively lower temperature and albedo can be studied using the model described in section 2.

Due to high humidity close to the water surface, there will be higher moisture ingress. But the lowered temperature over the water surface exponentially slows down all reactions (e.g. corrosion). Therefore, we hypothesize that the overall degradation of the FPVs maybe even slower compared to panels on land. In some cases, however, the lake temperatures may be higher than that on land—FPVs should not be installed in such locations due to degraded performance and poor reliability.

6. Conclusions

The PV community has made good progress in terms of developing experimental and numerical frameworks to analyze cell to module to farm yield and economics. For farms:

(a) numerical models accurately predict light collection and output energy for any given location and configuration under specific ambient conditions.
(b) The farm should be optimized in terms of LCOE—this step is made easier through the concept of essential LCOE (LCOE*) by using the module to land cost ratio M_L. Recent analysis shows, currently, bifacial single-axis tracking may have the lowest LCOE.
(c) The farm lifetime depends on the module internal degradation rate, and the yield over the lifetime depends on both the degradation and external losses (e.g. soiling). Statistical effects of such reliability factors can also be predicted using a combination of empirical and physics-based models. Although there have been only a few long term reliability studies on in-field bifacial modules, it has been seen that the degradation rate of these bifacial modules is comparable to that of the more mature monofacials.

Armed with these tools (e.g. PVsyst [8], NREL-PVWatts [69], NREL-SAM [70], Sandia-PVLib [71], Purdue (PUMET, PUB) models [72, 73], RdTools [45]) we can now design the next generation of bifacial PV systems, e.g. tracking, agro-photovoltaics, FPV systems, etc. Such compound systems are deemed promising as they give us the opportunities to more efficiently utilizing our natural resources.

The physics-based modeling tools are complicated to implement and computationally expensive to run. These may not be accessible for all, and it may be impossible to numerically calculate for all configurations and locations. There is an opportunity to redesign our modeling or analysis space—a machine learning based, physical parameter aware, extrapolation functions may be created from a finite set of already calculated results. This may be the final pathway towards the next generation compound PV system designs.

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